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Health System Management Group Assignment for Clinical Year II Students

Team 10 - Principles and components of health management information system and HMIS and the current DHIS2 in Ethiopia

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Principles and components of health management information system and HMIS and the current DHIS2 in Ethiopia

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I. Introduction

A well-functioning Health Management Information System (HMIS) is the cornerstone of evidence-based health planning, decision-making, and policy formulation. In many low- and middle-income countries, including Ethiopia, strengthening HMIS is essential for improving health service delivery, ensuring resource efficiency, and achieving national and global health goals. HMIS serves as an integrated platform for collecting, processing, analyzing, and disseminating health-related data to support effective health system management at all levels.

The principles guiding HMIS include data accuracy, completeness, timeliness, accessibility, confidentiality, and relevance. These principles ensure that information generated is reliable and usable for strategic decision-making. Components of HMIS typically include data collection tools and processes, indicators, data sources, data management systems, and feedback mechanisms. Together, these components enable the systematic flow of information from service delivery points to higher levels of the health system.

In recent years, Ethiopia has made significant strides in modernizing its HMIS through the adoption of the District Health Information Software 2 (DHIS2), a flexible, open-source platform designed for health data management. DHIS2 facilitates real-time data collection, analysis, and visualization, enhancing the country's capacity to monitor health service performance and outcomes. Implemented nationwide, the DHIS2 platform supports Ethiopia's Health Sector Transformation Plan (HSTP) by enabling more efficient reporting, improved data quality, and strengthened accountability in the health sector.

II. Guiding Principles of HMIS

1. **Standardization:** Use common definitions, tools, and procedures to ensure consistency and accuracy in data collection and analysis across all regions and time periods.
2. **Integration:** Develop one unified HMIS reporting system per administrative level, used by all stakeholders, to avoid duplication and promote data sharing.
3. **Simplification:** Focus only on collecting and analyzing data that directly improves performance, saving time and resources.
4. **Accessibility:** A logical and transparent structure is instrumental to enable the public and other stakeholders' access to information
5. **Flexibility:** HMIS needs to be flexible in order to adapt to epidemiological changes, the state of health of the population, emerging technology and needs.
6. **Data Quality:** Data needs to be accurate, complete, timely and reliable.
7. **Confidentiality:** HMS must maintain the confidentiality of patient data and limit access to authorized personnel

- 8. Innovation:** Encourage local-level innovations (from woredas, facilities, etc.) and replicate successful approaches across the system.
- 9. Partnership:** Leverage strong support from the Ministry of Health and partners, including the private sector.
- 10. Sustainability:** Ensure the system is affordable and easy to use. Minimize costs related to ICT, stationery, and labor, while meeting the needs at every administrative level.

III. Components of health management information system

- 1. Data Collection:** The process of gathering health-related data from various sources, such as: Health facilities, Community health workers, population Surveys and routine reporting systems
- 2. Data management:** The process of storing, organizing, and maintaining health-related data in a structured manner using databases, spreadsheets, and specialized health software. It ensures that collected data are reliable, secure, and readily available for use.
- 3. Data analysis:** This component transforms raw data into useful information by identifying trends, performance indicators, and problem areas. Analysis may be quantitative (e.g., rates, proportions) or qualitative.
- 4. Reporting:** The process of generating reports that communicate health-related information to health managers, policymakers and stakeholders. This may include tables, graphs, dashboards, or written reports.
- 5. Information Dissemination:** After reporting, information should be shared through appropriate channels to ensure that all stakeholders have access to actionable data. The channels can be websites, mobile applications or dashboards.
- 6. Monitoring and Evaluation:** The process of tracking the performance of the HMIS to ensure that it is meeting its objectives and making improvements where necessary.
- 7. Human Resources:** A skilled workforce is essential for running and maintaining HMIS. This includes data collectors, statisticians, IT staff, data clerks, and health managers. Continuous capacity-building and training are also vital.
- 8. Infrastructure:** The hardware and software systems required to support the HMIS, including servers, computers, power supply, network infrastructure and software applications used for data management and analysis.
- 9. Standards and Protocols:** The guidelines and protocols that govern the collection, management, and analysis of health-related data to ensure consistency, comparability and accuracy.

10. Funding and Sustainability: Sufficient and continuous funding is necessary for implementing and maintaining the HMIS infrastructure, software licenses, training, and staffing. Sustainability also involves government ownership and integration with national health priorities.

IV. HMIS in Ethiopia

In Ethiopia, prior to the implementation of the **HMIS**, only some locations practiced performance monitoring. The HMIS was not originally designed to provide standard and consistent sector-wide or program indicators. Data delivered through the HMIS come from service delivery and administrative records, maintained as part of routine transactions at health facilities and management offices. In a well-performing HMIS, data should be collected from every Health Institution (HI) in the country.

The HMIS was first piloted in Ethiopia in 2006/07 and has been implemented nationwide since 2008. Since then, the health system has shown significant improvement. HMIS is one of the seven components of the Health Sector Development Program (HSDP III). A functional HMIS and Monitoring & Evaluation (M&E) system are the backbone of effective health care delivery. Key elements for successful program management and implementation include designing a program built on a hierarchy of objectives, targets, activities, and measurable indicators. The objectives of M&E are to improve program management, ensure optimum use of resources, and enable timely decision-making to resolve constraints and implementation challenges.

HMIS/M&E Vision

To achieve continuous improvement in the health of the people of Ethiopia through effective use of information.

HMIS/M&E Mission

To support continuous improvement of health services and the population's health status through action-oriented, evidence-based decision-making based on quality information.

HMIS/M&E Goals

- To support decentralized, action-oriented, evidence-based decision-making.
- To provide health information to international, national, and decentralized authorities; elected officials; and other public and private sector organizations.

Five strategic issues have been identified as critical to strengthening and continuously improving health sector HMIS/M&E:

1. Capacity building
2. Standardized and integrated data collection and reporting
3. Linkages between information sources
4. Information use through action-oriented performance monitoring
5. Appropriate technology

In most health institutions in Ethiopia, the HMIS still does not deliver its most important product—quality information that supports monitoring and performance improvement. As a result, M&E remains weak, lacking the reliable data foundation that a strong HMIS should provide.

Current HMIS/M&E weaknesses include:

1. Incomplete institutionalization
2. Unstandardized data collection
3. Unintegrated reporting and data transmission
4. Weak information use (analysis and interpretation)
5. Limited resources for HMIS/M&E

The HMIS was specifically designed to address these challenges.

V. DHIS2 in Ethiopia

“DHIS2 revolutionizes data access, liberating us from HMIS constraints. Previously, accessing data was restricted by time and place. Now, with DHIS2, data is at our fingertips anytime, anywhere, contingent upon stable connectivity. It’s a transformative leap in accessibility and efficiency.” – Regional primary health care unit leader

DHIS2 which stands for district health information system two is an open source, web-based platform most commonly used as a health management information system (HMIS), but that is usable in any sector due to its flexible and customizable design and it is managed by the HISP Centre at the University of Oslo (UiO)

DHIS2 is a tool for collection, validation, analysis, and presentation of aggregate and patient-based statistical data that are routinely collected in all public health facilities in a country using the system

Each country (or organization) has its own “instance” of the DHIS2 platform with full ownership of the application and the data contained within it. A crucial aspect is that data ownership rests with each institution using the software. This implies that the general data and privacy regulations as well as the practical matters such as user permissions and system access administration.

Today, DHIS2 is the world’s largest HMIS platform, in use by 80 low and middle-income countries, making it the government-wide system of record for health data in these countries.

Ethiopia’s HMIS is one of the world’s largest DHIS2 implementations, and it has greatly strengthened Ethiopia’s capacity to collect, analyze, and act on health data.

Previously the country relied largely on paper-based registers and manual reporting of health data until the country began its transition to a hybrid paper-digital record keeping process in 2008.

From 2008 until 2017, health data was entered into two separate, desktop-based, proprietary HMIS systems operated in different regions, which had been developed with support from various organizations. This use of disparate systems created significant challenges in interoperability, dramatically limited data analysis and use and hampering informed decision making.

Additionally, the HMIS reporting forms were not standardized, and data for these two standalone systems was pulled from records maintained by health facilities and management offices, which made it quite difficult to compare data across different locations,



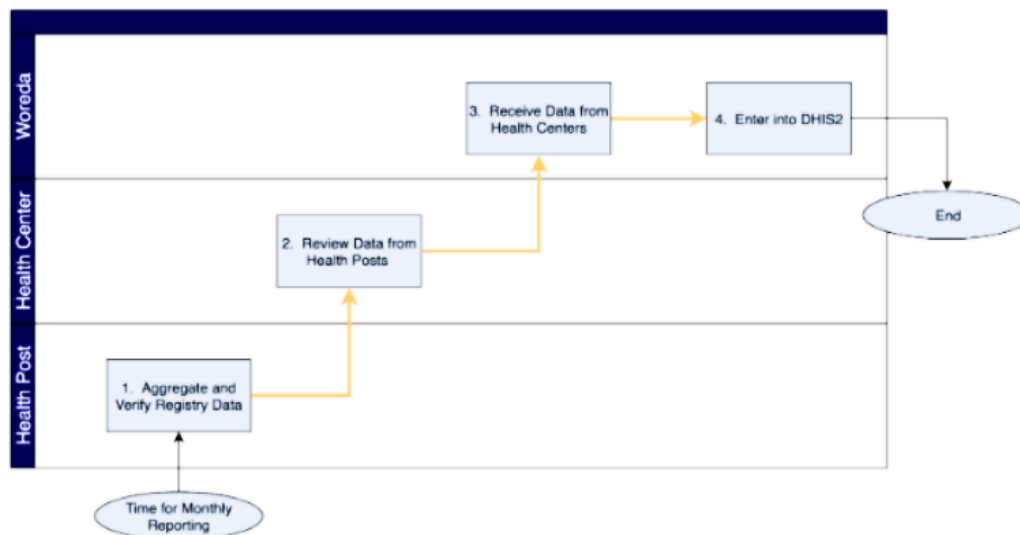
In 2017, the Ministry of Health began working toward full national implementation of DHIS2 as a health management information system (HMIS), to improve health data management in the country, and enable informed decisions about policies, interventions, and resource allocation. It has made impressive progress toward this goal over the last seven years, beginning with initial customization and deployment of DHIS2 in 2017, training more than 7,000 health workers and managers to use the software, and expanding data collection to cover all of the nation's 30,000+ public health facilities as of 2024, providing data on health services and outcomes for that serve a population of more than 120 million people.



More than 95% of government health facilities consistently reported through DHIS2 between 2018 and 2022, with more than 90% completeness and 70% timeliness on average. Digitizing health data with DHIS2 has drastically reduced reporting time in many areas, from up to 26 days using the previous process, to as few as five days using DHIS2.

The monthly reporting is generally expected from health facilities by day 21, with health posts sending data to health centers by day 23 and Health centers finalizing reports by day 26.

The DHIS 2 Implementation in Ethiopia was highly significant due to the large population size in our country. The rollout incorporated several local innovations as a result of the specific challenges encountered in each context.



However, the following problems have hindered effective use of the platform;

- Lack of efficient HIS and DHIS 2 training
- Inadequate analytical abilities
- Lack of internet access and electricity
- Availability of human resources

As a result, the data on the ground tells a slightly different story. The overall DHIS2 implementation in Ethiopia's public health facilities varies by region but generally falls within a good to very good range.

- **Dire Dawa** reports an 80% implementation level with strong infrastructure (80%) and very good data quality (87.5%), alongside 64% training coverage and 70.3% data utilization.
- **Oromia** leads with an 88.6% overall score, though with lower training (31.3%) and utilization (52.8%) rates.
- **Tigray** reaches 75.2% overall
- Other regions such as Amhara and Hadiya report lower training and utilization levels, around 12.8–35% and 22.5–69%, respectively.

VI. Summary

A Health Management Information System (HMIS) is an organized system designed to collect, store, manage, analyze, and disseminate health-related data. A well-organized HMIS is necessary for managing health services effectively.

In Ethiopia, HMIS plays a key role in collecting, processing, and using health data to guide policy decisions, planning, and resource allocation. The system is based on important principles such as standardization, accessibility, confidentiality, interoperability, flexibility, integration, data quality, timeliness, sustainability, and focusing on users' needs. These principles help ensure the information produced is trustworthy and useful for improving the health system.

The main components of HMIS include collecting data from various sources, managing that data in an organized way, analyzing it to find important trends, reporting the results clearly, sharing information efficiently, and regularly monitoring and evaluating the system's performance.

Ethiopia's HMIS started in 2006/07, evolving from fragmented manual processes to more cohesive data systems. Although there were early problems with inconsistent reporting, the country has made major progress by adopting national standards and focusing on capacity building and better information use.

The biggest recent improvement has been the countrywide rollout of the District Health Information System 2 (DHIS2), an open-source, web-based platform that was designed for low and middle income countries. Since 2017, Ethiopia has moved away from paper and separate digital systems to DHIS2. This system is being used by governmental health facilities (over 95%) non-governmental and other organizations. This digital shift improves efficiency and supports better decision-making at all levels. DHIS2's adaptability and open design also allows for customizations to meet specific needs of different regions and strengthens accountability and ongoing health sector monitoring, directly supporting Ethiopia's Health Sector Transformation Plan.

In summary, Ethiopia's HMIS, powered by DHIS2, shows how well-designed, technology-based systems can improve health data use, provide accurate and timely data, speed up information flow, and ultimately help the country make better health outcomes a reality for its people.

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